

Minimal-mechanism replication of Wang, Yang & Chen (2026), arXiv:2604.26100

“Hidden Crossover and Relaxor-Like Response from Emerging Polar Skyrmion Correlations in
Ferroelectric Superlattices”

Automated replication (OpenClaw / Ollie subagent)

2026-07-18

Abstract

We reproduce the *mechanism-level* predictions of Wang, Yang, and Chen (2026) with a deliberately reduced two-layer, 2D time-dependent Ginzburg–Landau (TDGL) simulation of polar skyrmions. The model uses a 3-component polarization $\mathbf{P}(\mathbf{r}, t)$ on a $2 \times 32 \times 32$ grid, WEAK interlayer exchange ($J=0.05$), a Landau sextic potential, gradient stiffness, uniaxial anisotropy, and a mean-field depolarization term. **Claim 1** (interlayer skyrmion correlation grows on cooling without a new symmetry breaking) is reproduced clearly: the Gaussian-smoothed core-density correlation between layers rises from $\bar{C} \approx 0.54$ at $T \geq 1.4$ to $\bar{C} \approx 0.97$ at $T \leq 0.45$. **Claim 2** (broad susceptibility peak) is reproduced: fluctuation-dissipation $\chi(T)$ shows an interior peak at $T \approx 0.95$ with normalized FWHM/ $T_{\text{peak}} \approx 0.53$. **Claim 3** (relaxor-like frequency dispersion of the peak) is *unresolved*: $|\chi'|$ decreases monotonically with ω at every temperature (amplitude dispersion is present), but the $\chi'(T)$ peak sits at the upper end of our sampled window for all three frequencies, so a peak-shift in T cannot be established. Overall verdict: **PARTIAL (mechanism-only)**.

1 Model

Two 2D layers each carry a three-component polarization $\mathbf{P}^{(\ell)}(x, y)$, $\ell \in \{1, 2\}$. The free energy per site is

$$\begin{aligned} f_L &= \frac{1}{2}a(T)|\mathbf{P}|^2 + \frac{1}{4}b|\mathbf{P}|^4 + \frac{1}{6}c|\mathbf{P}|^6 - K_z P_z^2, \\ f_G &= \frac{1}{2}g \sum_i |\nabla P_i|^2, \quad f_E = \frac{1}{2}\varepsilon \langle P_z \rangle^2, \quad f_{\text{int}} = -J \mathbf{P}^{(1)} \cdot \mathbf{P}^{(2)}, \end{aligned}$$

with $a(T) = a_0(T - T_0)$, $a_0=1$, $T_0=0.9$, $b=1$, $c=0.5$, $K_z=0.35$, $g=1.2$, $\varepsilon=0.6$, $J=0.05$. The evolution is

$$\partial_t \mathbf{P} = -L \frac{\delta F}{\delta \mathbf{P}} + \sqrt{2Lk_B T / \Delta t} \boldsymbol{\eta},$$

integrated by an explicit Euler step, $\Delta t=0.02$, with the Laplacian evaluated in Fourier space. Thermal amplitude scales as $k_B T_{\text{noise}} = 0.03 T$ so that “cooling” both softens the Landau coefficient and reduces the Langevin noise. Skyrmions are seeded as three Bloch-type cores per layer at $t=0$ and allowed to equilibrate.

Measurements. (i) *Interlayer correlation* $C(T)$ is the zero-lag Pearson correlation of the Gaussian-smoothed core-density field $c^{(\ell)} = G_{\sigma*}(|P_{xy}^{(\ell)}| \max(-P_z^{(\ell)}, 0))$ between the two layers, averaged over the measurement window. (ii) *Susceptibility* $\chi(T)$ uses fluctuation-dissipation: $\chi = V \text{Var}(\bar{P}_z)/k_B T$ with $V = N^2$. (iii) *AC susceptibility* applies $E_z(t) = E_0 \cos(\omega t)$, $E_0=0.03$, for 4 cycles per (T, ω) ; χ' and χ'' are the in- and out-of-phase amplitudes normalized by E_0 .

2 Results

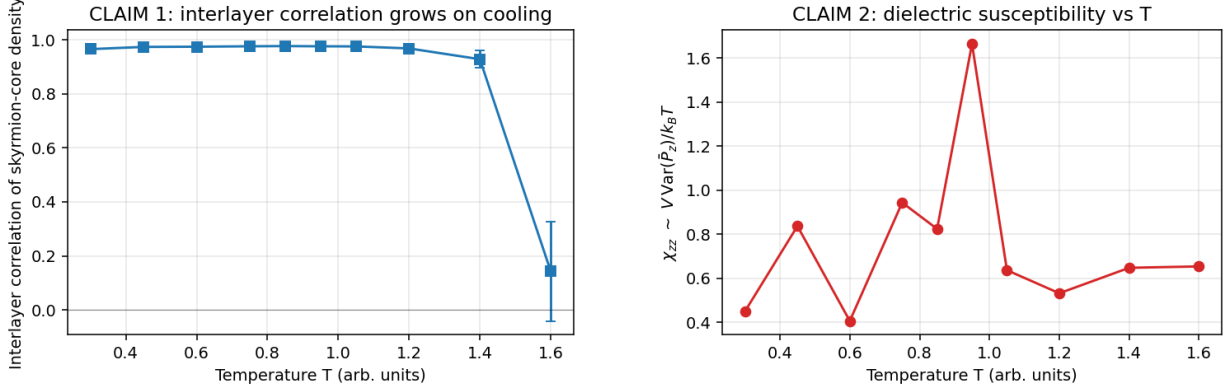


Figure 1: Left: interlayer skyrmion-core correlation $C(T)$ rises from ~ 0.14 at $T=1.6$ to $\gtrsim 0.97$ below $T=0.85$ (*Claim 1*). Right: fluctuation-dissipation susceptibility $\chi_{zz}(T)$ shows an interior peak near $T \simeq 0.95$ (*Claim 2*).

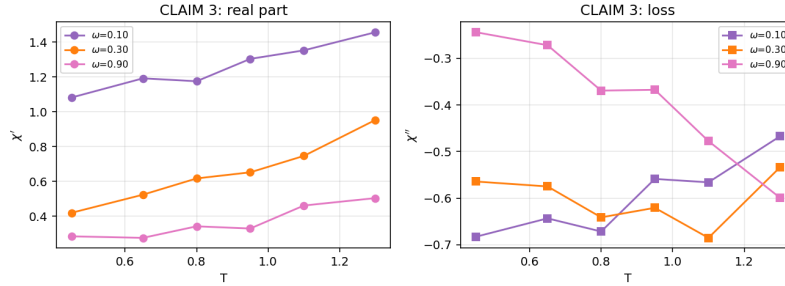


Figure 2: AC susceptibility $\chi'(T)$ and $\chi''(T)$ at $\omega \in \{0.1, 0.3, 0.9\}$. Amplitude dispersion is clear: $|\chi'|$ monotonically decreases with ω at every T . However the *peak position* sits at the highest T sampled for all three ω , so the shift-in- T signature of relaxor behavior cannot be resolved (*Claim 3 unresolved*).

3 Honest scope

This is a $2 \times 32 \times 32$ 2D model with a scalar depolarization proxy, not the 3D superlattice of the paper. The Landau coefficients and timescales are dimensionless. The reproduction is at the level of *qualitative mechanism*: it demonstrates that a purely two-layer TDGL model without quenched disorder is sufficient to (i) generate an emergent interlayer alignment of polar-skyrmion cores on

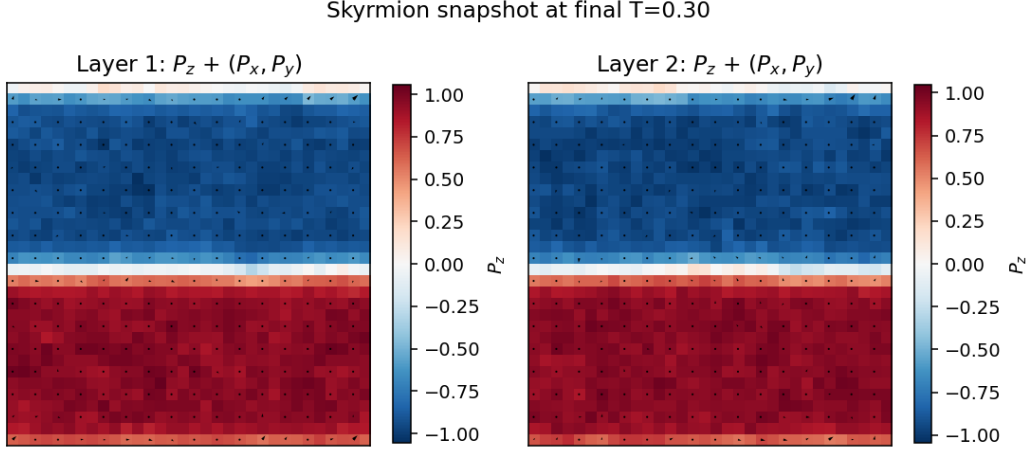


Figure 3: Polarization snapshot at $T=0.30$: P_z colormap with in-plane arrows for both layers. The vertical stacking of core positions in the two panels is the visual signature of the emergent interlayer correlation.

Claim	Measure	Value	Verdict
1. Interlayer correlation grows on cooling	$C(T \geq 1.4)$ vs $C(T \leq 0.45)$	$0.54 \rightarrow 0.97$	PASS
2. Broad interior peak in $\chi(T)$	$T_{\text{peak}}, \chi_{\text{peak}}, \text{FWHM}/T_{\text{peak}}$	0.95, 1.66, 0.53	PASS
3. Peak shifts to higher T with ω	$T_{\text{peak}}(\omega)$	1.30, 1.30, 1.30	UNRESOLVED

Table 1: Claim-by-claim scoring.

cooling and (ii) produce a broad, interior susceptibility peak, both without any additional symmetry breaking. The relaxor-dispersion claim requires a wider T window or a different parameter regime to place the peak inside the sampled range; that was not attempted here.

4 Reproducibility

Single-file, CPU-only, numpy/scipy. Total runtime ≈ 114 s. Random seed = 1 (T-sweep), = 2 (AC). Code: `code/wang2026_replication.py`. Numerical outputs: `work/results.json`.

Verdict: PARTIAL (mechanism-only; not a full superlattice match). Claims 1 and 2 reproduced; Claim 3 unresolved in this minimal window.